

TOPIC 8.A

Structure, Function, and Health of the Musculoskeletal System

1. List the main components of the musculoskeletal system.
2. How are bones and muscles connected to each other?
3. What are the components of human bone, and how does their ratio change with age?
4. What are the most common disorders of the musculoskeletal system, and what are their causes?
5. In your opinion, what are the key aspects of prevention and correction? What lifestyle advice would you give to your peers?

Expected Knowledge & Marking Scheme

	Expected Knowledge	Attainable score	Attained score
1.	Passive: bones – support the body, protect internal organs, participate in movement. Active: muscles – attached to the bones to move them.	1+1 pt	
2.	Bones can be connected to each other movably via joints (e.g., wrist) and immovably via sutures (skull), synostosis/fusion (pelvic bone), or intervertebral discs.	3 pts	
3.	Inorganic matter: $Ca_3(PO_4)$, $CaCO_3$, $MgCO_3$; these provide the rigidity of the bones. Organic matter: protein filaments, connective tissue fibres; these provide elasticity. Ageing: In youth, bones contain more organic matter, making them more flexible; in old age, there is more inorganic matter, making bones more brittle and fractures harder to heal.	3+3+3 pts	
4.	Congenital disorders: Hip dysplasia – the femoral head slips out of place. Acquired disorders: Flat feet (pes planus), scoliosis, osteoporosis (typical in older age). Accidental injuries: Strain; sprain (the bone moves out of place but returns to its original position); dislocation (moves out of place and remains fixed there).	2+2+2 pts	
5.	Hip dysplasia: Early screening (in infancy); treatment: bracing or	2+2+4+2 pts	

	Expected Knowledge	Attainable score	Attained score
	surgery. Flat feet: Suitable footwear, strengthening foot muscles, avoiding obesity. Scoliosis: Currently almost a widespread disease. Cause: poor posture, uneven or excessive loading of the back. Prevention: avoiding the above, using two-shoulder backpacks, strengthening trunk muscles through regular exercise/swimming. Correction: remedial gymnastics, surgery in severe cases. Osteoporosis: Risk increases with age. Prevention: consuming calcium-rich foods, avoiding excessive coffee, tea, and soft drinks (as these cause calcium loss), and regular physical activity.		
	Structure of the response, expressive skills	5 pts	
	Total	30 pts	

TOPIC 8.B

Procedures during a Smog Alert

On a hot summer day, the Mayor of Budapest has ordered a smog alert due to high air pollution lasting for days. In the capital, the concentration of air pollutants measured simultaneously at three monitoring stations exceeded the "information threshold."

1. Define what smog is.
2. What types are there, and what are their causes?
3. What actions must be taken in the case described above?

Expected Knowledge & Marking Scheme

	Expected Knowledge	Attainable points	Attained points
1.	If front-free, windless weather persists in a large city or industrial area, the concentration of soot and particulate matter (PM) can accumulate to dangerous levels. Interacting with weather conditions, these create smog (smoke-fog) filled with corrosive compounds that are severely harmful to health.	5 pts	
2.	Summer or Los Angeles-type smog: pollutants accumulating from exhaust fumes (NO_2, CO). Winter or London-type smog: mainly caused by CO and SO_2 released during the combustion of fossil fuels. The latter is responsible for the formation of acid rain.	5 pts	
3.	It is advisable to wear a mask; burning garden waste is prohibited; outdoor sports should be avoided. Do not ventilate extensively; stay outdoors as little as possible. Three levels of thresholds: a) Health limit value: exceeding this endangers health. b) Information threshold: poses a danger to sensitive groups (children, the elderly). Information and calls to action are issued via media so that healthy individuals also avoid outdoor sports and activities involving intense breathing. c) Alarm threshold: reaching this endangers every city resident staying outdoors; therefore, the municipality	10 pts	

	restricts car traffic and open-air burning, may shut down polluting plants, and may temporarily make public transport free.		
Total		20 pts	

